

Chapter 21

Inhibitory control and voluntary tic suppression

SIMON MORAND-BEAULIEU^{1,2,*}

¹*Department of Psychology, McGill University, Montréal, QC, Canada*

²*Department of Psychology, Université du Québec à Montréal, Montréal, QC, Canada*

Abstract

Tics, such as those seen in Tourette syndrome (TS), are brief, sudden, and involuntary movements or vocalizations. Although tics are involuntary, most individuals with TS can suppress them on command for short periods of time. This capacity, which is called voluntary tic suppression, is a hallmark feature of TS and differentiates it from most other movement disorders. Several studies have investigated inhibitory control of voluntary actions in TS, given its potential associations with both tic expression and voluntary tic suppression. However, studies on response inhibition in TS have produced quite mixed results. Furthermore, the mechanisms of voluntary tic suppression remain poorly understood. The current chapter offers an overview of the evidence related to inhibitory control of voluntary actions and voluntary tic suppression in TS. It reviews evidence from clinical, behavioral, neuropsychologic, and neurobiologic studies in order to understand to what extent these processes may be similar or different.

INTRODUCTION

The study of Tourette syndrome (TS) is quite fascinating. Though it is fundamentally a neurologic disorder, its severity is influenced by several factors that are not directly related with its neurologic origin. Fatigue, boredom, anxiety, thinking about or focusing on tics, and social situations are all factors that have been shown to increase tic severity (Iverson and Black, 2022). Furthermore, tics change across time, and have been shown to have a fractal distribution (Peterson and Leckman, 1998; Beeler et al., 2022). In essence, within an hour, there can be a few minutes where tics will be more intense. Within a day, there will likely be a few hours with more severe tics. During a week, there will for sure be days when tics seem more present than others. Same goes for months, years, and even decades. Thus, tic frequency and severity are highly variable and influenced by several factors.

Tics are often preceded or accompanied by feelings of discomfort, tension, or urge to move, which are known as premonitory urges (Cohen and Leckman, 1992; Leckman et al., 1993; Cavanna et al., 2017). These urges can be felt as focal or generalized uncomfortable sensations (Rajagopal et al., 2013). Discomfort or tension can often be temporarily relieved following the expression of a single or a series of multiple tics (Leckman et al., 2022). Premonitory urges have been recognized

as a crucial feature of TS for a long time. In the early 20th century, Meige and Feindel (1902) reported the existence of an irresistible impulse preceding tic onset. An important description of premonitory urges was later given by Joseph Bliss, who reported various unpleasant sensations in certain parts of his body prompting him to tic in order to get relief from those sensations (Bliss et al., 1980). Such reports revealed that tics were not as involuntary as once thought. Tics have thus sometimes been described as “unvoluntary” (Tourette Syndrome Classification Study Group, 1993), because they can be considered as being halfway between voluntary and involuntary movements (Cavanna and Nani, 2013). Thus, if tics are not fully involuntary movements, they can be controlled to an extent. Indeed, in most individuals with TS, tics can be suppressed for short periods (Ueda et al., 2021). Voluntary tic suppression is a core feature of TS and has been the subject of several investigations, given its relevance as a mechanism leading to decreased tic severity. Researchers interested in tics and their suppression often used tasks assessing the voluntary control of actions to better understand the neurocognitive and neurobiologic correlates of voluntary tic suppression. Thus, response inhibition became one of the most studied neurocognitive processes in TS (Morand-Beaulieu et al., 2017b). Over the years, studies of volitional inhibitory control have shared different interpretations about tics and their suppression. Such views notably include that (1) tics

*Correspondence to: Simon Morand-Beaulieu, PhD, Department of Psychology, Université du Québec à Montréal, 100 Sherbrooke W Street, Montréal, QC H2X 3P2, Canada. E-mail: morand-beaulieu.simon@courrier.uqam.ca.

(and the inability to control them) could result from impairments in volitional inhibitory control and (2) neural processes involved in response inhibition during experimental tasks may also be involved in tic suppression. The former view is considered to be somewhat outdated nowadays (see [Ganos et al., 2018b](#); [Jackson and Jackson, 2020](#)), and the latter has been the subject of several recent studies (e.g., [Rae et al., 2020](#)).

This chapter will first review evidence from neuropsychologic and neurobiologic studies pertaining to response inhibition in children and adults with TS. The focus will then shift to studies pertaining to clinical, cognitive, and neurobiologic features of voluntary tic suppression. The final section of the chapter focuses on whether there are shared features or mechanisms between response inhibition and voluntary tic suppression.

RESPONSE INHIBITION IN TOURETTE SYNDROME

In TS, behavioral assessment of response inhibition using neuropsychologic testing mostly began in the 1990s (e.g., [Ozonoff et al., 1994](#); [Ozonoff et al., 1998](#); [Sherman et al., 1998](#)), and picked up steam in the next two decades. Those studies have provided quite mixed results, as diminished (e.g., [Goudriaan et al., 2005](#); [Eddy et al., 2012](#)), intact (e.g., [Marsh et al., 2007](#); [Roessner et al., 2008](#)), and even enhanced (e.g., [Jackson et al., 2007](#); [Thibault et al., 2009](#); [Jackson et al., 2011](#)) response inhibition abilities have been reported. Several factors, such as the presence of co-occurring disorders, medication intake, or the use of different experimental tasks, may explain such differences across studies. A meta-analysis, which included 61 studies and more than 1700 individuals with TS, aimed to better understand the portrait of response inhibition in TS ([Morand-Beaulieu et al., 2017a](#)). The main finding was that both children and adults with TS showed reduced response inhibition abilities compared to individuals without TS ([Fig. 21.1](#)). Between-groups differences were evident for sentence completion tasks, Stroop tasks, circle tracing tasks, and the Continuous Performance Test (CPT). However, for other commonly used tasks, such as the Go/NoGo task or the stop-signal task, there were no significant between-group differences. Attention-deficit/hyperactivity disorder (ADHD) was found to have an important impact on response inhibition. Effects sizes for between-group comparisons with controls were twice as large in individuals with TS+ADHD, compared with those without co-occurring ADHD. However, reduced performance in response inhibition tasks was also found in individuals without clinically diagnosed co-occurring conditions. Those differences were small but could suggest that there are some difficulties with response inhibition that are inherently part of TS. However, even without a diagnosis of a co-occurring condition, some traits associated with common co-occurring conditions, such as inattention, hyperactivity, or obsessive-compulsive symptoms, may have been present and may be responsible for such differences. For instance, many studies using a 2x2 factorial design to assess the co-occurrence of TS

and ADHD have reported enhanced inattention/hyperactivity symptoms in children with TS without ADHD relative to typically developing controls ([Shephard et al., 2016](#); [Openneer et al., 2020](#); [Morand-Beaulieu et al., 2022](#)). Thus, it is hard, in experimental studies and even more in meta-analyses, to disentangle the impact of the core symptoms of TS (i.e., tics) on neuropsychologic measures of response inhibition from that of other frequent co-occurring symptoms, such as hyperactivity or inattention.

Since the publication of that meta-analysis, response inhibition continues to be an important topic in TS research. Notably, recent studies have tried to parse out distinct aspects of cognitive control to better understand how it may be related to the pathophysiology of TS. For instance, a few studies have investigated automatic inhibition with the aid of subliminal priming. In this type of paradigm, a subliminal prime (or cue) is presented, which is followed by a supraliminal target after a short delay. That target can be either compatible or incompatible with the prime. The delay between the prime and the target is called the stimulus onset asynchrony. A short (<100 ms) or very long (>200 ms) stimulus onset asynchrony usually improves performance (positive compatibility effect), while stimulus onset asynchronies between 100 and 200 ms typically decrease performance (negative compatibility effect). The negative compatibility effect is thought to reflect the automatic inhibition of motor responses ([Eimer and Schlaghecken, 2003](#); [Schlaghecken et al., 2008](#)). Using such paradigm, [Stenner et al. \(2018\)](#) found intact automatic inhibition in adults with TS. However, they only assessed automatic inhibition at a single stimulus onset asynchrony, which does not allow investigating the full range of priming effects. This issue was notably tackled by [Rawji et al. \(2020\)](#), who used a masked priming task with eight different stimulus onset asynchronies ranging from 0 to 250 ms. In their study, they also used a stop-signal task in order to assess both volitional and automatic inhibition in the same sample of adults with TS. By using those two tasks, they wished to investigate processes associated with not only tic suppression but also tic generation. They found intact volitional inhibition (including both reactive and proactive inhibition) of motor responses in TS, as assessed with the stop-signal task. However, in the masked priming task, adults with TS did not show a negative compatibility effect at stimulus onset asynchronies of 100 or 150 ms, which suggests impaired automatic inhibition in those individuals. Based on those results, the authors suggested that intact volitional inhibition could be used to voluntarily suppress tics, while deficits in automatic inhibition may give rise to tics. In another study, proactive and reactive inhibition in TS were further parsed using advanced metrics during the stop-signal task ([Indrajeet et al., 2022](#)). The metric representing proactive inhibition estimated the time spent waiting for a stop signal in order to improve the chances of successfully inhibiting the motor response. The metric representing reactive inhibition estimated the slowing down of preparatory activity initiated by the onset of the stop signal. Using those advanced metrics, the authors found that only reactive inhibition was impaired in adults with TS, while proactive inhibition was intact. These

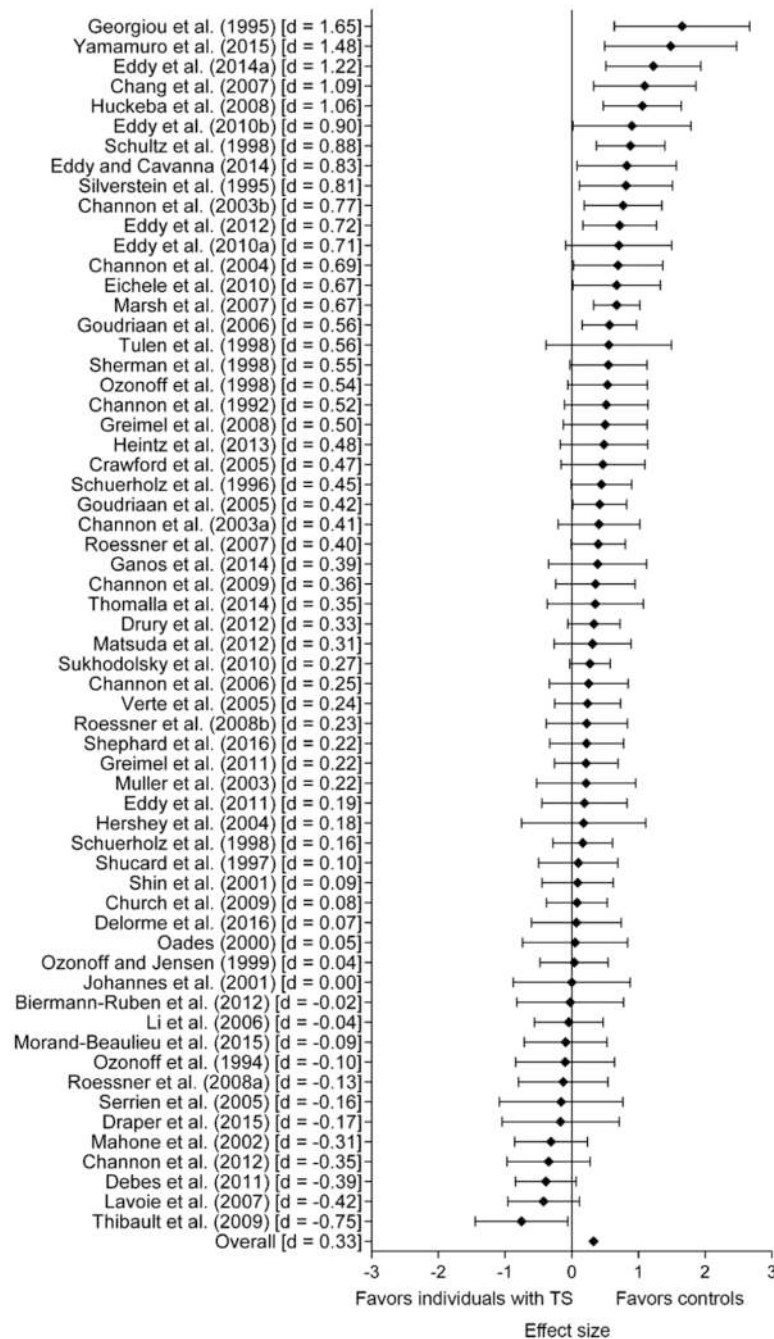


Fig. 21.1. Meta-analysis of response inhibition in TS. Individuals with TS had more inhibitory difficulties during response inhibition tasks than controls. Errors bars represent the 95% confidence interval and diamonds represent effect sizes from individual studies. *d*, Cohen's *d*; TS, Tourette syndrome. Adapted from Morand-Beaulieu S, Grot S, Lavoie J, et al. (2017a) The puzzling question of inhibitory control in Tourette syndrome: a meta-analysis. *Neurosci Biobehav Rev* 80: 240–262.

results suggest that individuals with TS do not differ from controls in their ability to postpone actions, but that the deceleration of the already initiated response appears to be reduced in participants with TS, relative to controls.

Several studies also assessed brain correlates of response inhibition in TS. Several brain regions, especially in the frontal cortex, are involved in inhibitory control. Notably, the right inferior frontal gyrus plays a major role in inhibitory control

and is part, along with the presupplementary area and the subthalamic nucleus, of a cognitive control network (Aron et al., 2007a, 2007b; Aron et al., 2014; Aron et al., 2016; Obeso et al., 2013). An common occurrence in the TS neuroscience literature is a pattern of enhanced frontal activation (Johannes et al., 2001; Thibault et al., 2009; Jackson et al., 2011; Morand-Beaulieu et al., 2018; Rae et al., 2020) as revealed by electroencephalographic (EEG) (see Fig. 21.2) and functional

magnetic resonance imaging (fMRI) studies, as well as increased functional connectivity in frontal regions (Serrien et al., 2005) for stimuli requiring volitional inhibitory control. This pattern has been suggested to represent compensatory processes implemented to maintain task performance, and possibly to regulate tic severity (i.e., through voluntary tic suppression; Marsh et al., 2007). Other recent studies investigating connectivity have highlighted important contributions of prefrontal regions to volitional inhibitory control in TS. For instance, Jurgiel et al. (2021) found increased effective connectivity between nodes of the fronto-parietal network in children with TS for incongruent trial of a Flanker task, which were associated with worse inhibitory performance. Similarly, worse inhibitory performance in a stop-signal task has been associated with increased connectivity between the right inferior frontal gyrus and the bilateral superior temporal gyri (Atkinson-Clement et al., 2020). That latter study also found interesting results pertaining to brain structure suggesting that worse inhibitory performance is correlated with increased gray matter signal in the inferior frontal gyrus and supplementary motor area.

Whether those increased activations and enhanced connections involving prefrontal regions represent compensatory

processes allowing to maintain good inhibitory performance and possibly control tics has been debated. Jung et al. (2013) have suggested that frontal overactivations may not constitute the source of inhibitory task-control signals but would rather lead to hyperexcitability of motor areas. Relatedly, several studies have investigated how motor areas contribute to volitional inhibitory control as well as the emergence of tics. Notably, the supplementary motor area appears to be an important structure in this context. For instance, Ganos et al. (2014c) found that increased activation in the supplementary motor area during action inhibition was positively correlated with tic severity. Those results suggest that the supplementary motor area could serve as a global inhibition mechanism involved in voluntary tic suppression and action inhibition. Furthermore, the lateralized readiness potential, an event-related potential partly generated by the supplementary motor area and associated with movement preparation and execution, was found to be larger in adults with TS and correlated with increased phonic tic severity (Morand-Beaulieu et al., 2018). Also, similar to their findings regarding the inferior frontal gyrus, Atkinson-Clement et al. (2020) found worse inhibitory performance to be associated with increased gray matter signal in the supplementary motor area.

There are also interesting findings pertaining to the role of the presupplementary in response inhibition. For instance, in a recent study, Rae et al. (2023) compared internally and externally cued action and inhibition in adults with TS. They used a technique called representational similarity analyses, which allows assessing the degree of similarity in neural responses to different task conditions. In the presupplementary motor area, they found increased similarity in adults with TS relative to controls for several contrasts: externally cued action and externally cued inhibition, internally cued action and internally cued inhibition, externally cued action and internally cued inhibition. However, only the last contrast remained significant after correction for multiple comparisons. These results reveal that adults with TS may have decreased differentiation between action and inhibition in the presupplementary motor area. The authors also mentioned that this decreased differentiation could explain why increased prefrontal activation is required to inhibit actions in TS.

The aforementioned findings from neurobiologic studies highlight several brain processes involved in the inhibition of volitional actions in individuals with TS. Yet, volitional actions differ from tics in several aspects. While tics may be controlled for short period via voluntary suppression, it remains unclear whether the same processes are involved in the inhibition of voluntary actions and tics.

VOLUNTARY TIC SUPPRESSION

Phenomenology of voluntary tic suppression

As mentioned above, voluntary tic suppression is a core feature of TS. That feature distinguishes it from most other movement

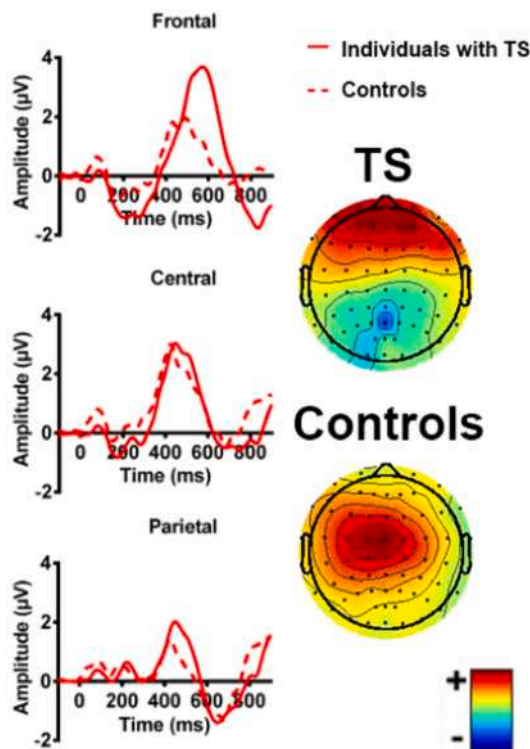


Fig. 21.2. Frontal shift of the NoGo P300 in TS. For stimuli requiring to inhibit a motor response (NoGo stimuli), adults with TS were found to have a frontal shift of the NoGo P300. In controls, the NoGo P300 was maximal over central electrodes. *ms*, milliseconds; *TS*, Tourette syndrome; μV , microvolts. Adapted from Morand-Beaulieu S, O'Connor KP, Blanchet PJ et al. (2018). Electrophysiological predictors of cognitive-behavioral therapy outcome in tic disorders. *J Psychiatr Res* 105: 113–122.

disorders (Ueda et al., 2021). A vast majority of individuals with TS are able to suppress their tics (Conelea et al., 2018), which however comes with the cost of a significant effort (Ganos et al., 2018b; Ueda et al., 2021). The ability to suppress tics does not seem to be influenced by co-occurring disorders such as ADHD or OCD, nor by psychotropic medication intake (Conelea et al., 2018). Voluntary tic suppression mostly occurs in situations where people dislike expressing their tics, such with unfamiliar people, at work or school, or in public places (Matsuda et al., 2016). Tic suppression abilities may be present from the very onset of tics (Greene et al., 2015) and are known to increase throughout childhood and adolescence (Banaschewski et al., 2003; Conelea et al., 2018). Voluntary tic suppression is also an important factor to consider in the developmental trajectory of TS, as reduced tic suppression abilities have been shown to predict worse tic severity one year later (Kim et al., 2019). There is a common idea that voluntary tic suppression leads to a rebound effect (i.e., an increase in tic severity following suppression), even among physicians and psychologists treating individuals with TS (Marcks et al., 2004). However, this idea is not supported by the scientific literature (Himle and Woods, 2005; Meidinger et al., 2005; Verdellen et al., 2007; Specht et al., 2013).

A better understanding of tic suppression has potential implications for treatments. However, voluntary tic suppression in itself must not be seen as a viable long-term option for coping with tics. Indeed, almost all individuals with TS report feeling uncomfortable while suppressing their tics (Matsuda et al., 2016). As pointed out by Ganos et al. (2018b), it is important to distinguish voluntary tic suppression from other types of tic control, such as control through behavioral, cognitive, or pharmacologic interventions. Tics can also be reduced through the modulation of attention (i.e., some people tic less when they are concentrated) or with aging/brain maturation. For instance, tics have been shown to be reduced when someone is with strangers or a physician, and to be increased when they are home alone (Goetz et al., 2001). Such diminished tic severity in specific contexts does not necessarily equate with voluntary tic suppression, as there may not be a conscious effort to suppress tics. Although those types of tic control share some overlapping features with voluntary tic suppression, there may be specific and distinct mechanisms underlying voluntary tic suppression.

Initial reports of volitional control over the expression of tics suggested that this ability was directly related to premonitory urges (Bliss et al., 1980). However, more recent research pertaining to the relationship between premonitory urges and the ability to voluntarily suppress tics has resulted in mixed findings (Ganos et al., 2018b), suggesting that this relationship is not straightforward. Premonitory urges most likely consist of a very complex phenomenon with important variance across individuals, which might explain some discrepancies across studies. Recently, Matsuda et al. (2020) developed a new scale to identify additional aspects of tic-related sensations: rumination about sensations and awareness of emerging tics. They reported that awareness was positively associated with voluntary tic suppression abilities, suggesting that awareness of emerging signals for tics facilitates self-initiated tic suppression.

Another interesting aspect of tic suppression is its somatotopy. Indeed, prior research has shown that tic suppression, just like tic expression, follows a somatotopic pattern (Ganos et al., 2015). Tic expression is more common and more severe in body parts with the largest cortico-striatal representations, such as the face or the hands. Conversely, in body parts that are less affected by tics, which typically have smaller cortico-striatal representation, voluntary tic suppression is more effective. Thus, this suggests that tics with a stronger motor command are harder to suppress (Ganos et al., 2018b). A similar pattern has been observed with median nerve stimulation, which may have stronger effects on more complex and peripheral tics, while more simple facial tics (e.g., blinking, mouth tics, etc.) may persist during stimulation (Morera Maiquez et al., 2020). These results must however be interpreted with caution, given the novelty to this treatment as well as a poor understanding of the underlying mechanisms (Iverson et al., 2023). Finally, the somatotopy of tic suppression is also consistent with the idea of selective suppression. For instance, in some settings, like a classroom, phonic tics can be selectively suppressed while some motor tics may persist (Ganos et al., 2015). Prior research has also shown that some individuals may have specific tics that are only visible at home when they are alone and not in other contexts (Goetz et al., 2001). However, it is not clear whether such type of selective tic suppression is really voluntary and conscious, or if some individuals unconsciously express specific tics only in certain settings.

Cognitive aspects of voluntary tic suppression

For those without tics, it is hard to understand what goes through the mind of people with TS in order to voluntarily suppress their tics. As mentioned by Ganos et al. (2018b), the processes involved in tic suppression can only be studied indirectly. It is however clear that this process requires an important and conscious effort (Matsuda et al., 2016). While the neurocognitive mechanisms underlying the initiation of voluntary tic suppression remain unclear, some studies have pointed toward specific cognitive functions associated with tic suppression. For instance, Specht et al. (2014) reported that verbal intelligence was correlated with the ability to suppress tics. Also, accuracy on tasks requiring attention or concentration has been shown to decrease during voluntary tic suppression (Conelea and Woods, 2008). Similarly, tic suppression is more efficient when people can focus on their tics, and less efficient when they need to divide their attention with external tasks (Herrmann et al., 2019). Those results are consistent with reports that tic suppression interferes with concentration (Wadman et al., 2016). It has been suggested that focusing on controlling tics makes it harder to give full attention to the external world (Cutler et al., 2009). Thus, if attentional and executive resources are required to effectively suppress tics, it is possible that those resources allocated to tic suppression are not available for other tasks. Alternatively, it is possible that the mechanisms that are involved and/or activated during voluntary tic suppression are incompatible with those required during tasks involving attention or executive control.

Neurobiologic correlates of voluntary tic suppression

Only a few studies have assessed the neurobiologic substrates of tic suppression. The first of such studies was conducted by [Peterson et al. \(1998\)](#), who contrasted fMRI activations during intervals of tic suppression and intervals of free ticcing. During tic suppression, they found both increased and decreased activity in various brain regions compared to intervals when participants were allowed to tic freely. Most subcortical areas showed reduced activation during tic suppression, except for the right caudate nucleus, which exhibited increased activity. Across the cortex, heightened activation was found in the right frontal cortex, bilateral temporal gyri, right anterior cingulate cortex, and bilateral occipital cortices, while decreased activation occurred in the right posterior cingulate cortex, bilateral hippocampus, cuneus, and left sensorimotor and inferior parietal cortices. That study, and in particular the increased right frontal activation, is often cited to support the idea that tic suppression recruits prefrontal brain regions that are also involved in response inhibition. This notion was further strengthened by further studies published afterward. Notably, frontal cortex activations during tic suppression were reported by two other fMRI studies in adults with TS. [Ganos et al. \(2014a\)](#) found increased regional homogeneity of the left inferior frontal gyrus during tic suppression relative to free ticcing. In another study, tic suppression mainly activated the superior prefrontal gyrus, the anterior and dorsolateral prefrontal cortex, and the dorsal anterior cingulate cortex ([van der Salm et al., 2018](#)). An EEG study investigating coherence between pairs of EEG electrodes as a proxy for functional connectivity revealed increased coherence in the alpha band over four pairs of frontal and central electrodes in nine adults with TS ([Serrien et al., 2005](#)). Enhanced coherence between those electrode pairs has been interpreted as increased functional connectivity between sensorimotor areas and the prefrontal and mesial frontal cortex. Similarly, a study by [Hong et al. \(2013\)](#) analyzed effective connectivity in nine children with TS, reporting increased coherence in the theta and beta bands during tic suppression. However, those two studies assessed functional connectivity between pairs of electrodes, which does not necessarily represent the functional connectivity between brain regions where those electrodes are located, due to volume conduction. This issue was tackled in a recent study assessing functional connectivity during tic suppression in the source space and in a large group of children with TS ([Morand-Beaulieu et al., 2023b](#)). Increased alpha-band functional connectivity was found during suppression (relative to free ticcing), but the brain regions involved in those connections somewhat differed from previous findings. There were three main hubs in the subnetwork showing increased functional connectivity during tic suppression: the right superior frontal gyrus, left precuneus, and left banks of the superior temporal sulcus ([Fig. 21.3](#)). Furthermore, and consistent with the idea that understanding tic suppression is relevant for behavioral treatments, functional connectivity within that subnetwork predicted reductions in vocal tic severity following comprehensive behavioral intervention for tics (CBIT) in children ([Morand-Beaulieu et al., 2023a](#)). All in all, studies assessing brain

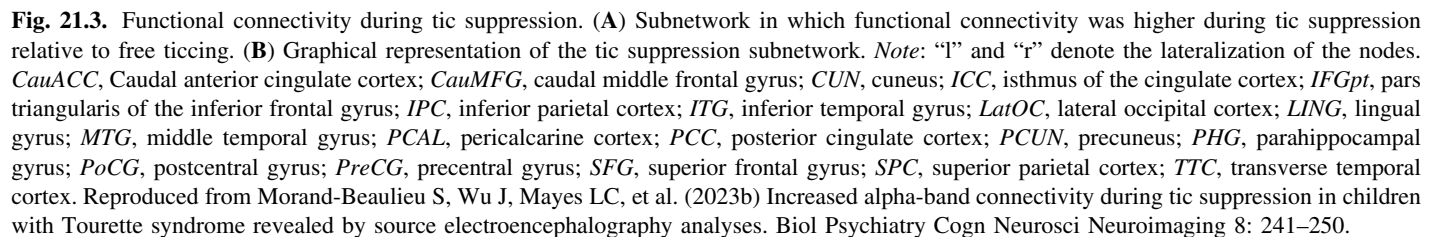
activation and functional connectivity during tic suppression revealed mixed results, with different and distributed brain areas involved in tic suppression. However, the involvement of prefrontal brain regions appears to be somewhat consistent across studies. Furthermore, there has been one study assessing the association between structural brain alterations in TS and voluntary tic suppression ([Ganos et al., 2014b](#)). Although gray matter volumes were reduced in two prefrontal structures, namely the right inferior frontal gyrus and the left frontal pole, those volume reductions were not associated with the capacity to voluntarily suppress tics.

Additionally, transcranial magnetic stimulation studies may provide interesting insights into the mechanisms of voluntary tic suppression. In a study assessing the effect of voluntary tic suppression on primary motor cortex excitability, it was discovered that voluntary tic suppression reduced corticospinal excitability, relative to free ticcing ([Ganos et al., 2018a](#)). Also, participants who were better at suppressing their tics showed larger differences in corticospinal excitability between voluntary tic suppression and free ticcing. These results, which inform us about how tic suppression modulates primary motor cortex excitability, are consistent with prior findings showing associations between reduced excitability of that region and decreased tic severity ([Orth et al., 2008](#)). Another study assessed the presupplementary motor area-mediated inhibition of the left motor cortex in children with TS ([Bruce et al., 2021](#)). It was found that less inhibition (of the left motor cortex) was associated with worse self-reported tic suppressibility, highlighting the importance of the presupplementary motor area in motor control in TS.

Finally, while the aforementioned studies investigated neural correlates of voluntary tic suppression during resting-state sessions, one study assessed the modulation of neural oscillations associated with voluntary tic suppression during the execution of a motor task ([Zapparoli et al., 2019](#)). In that study, the authors aimed to assess how tic suppression impacted the activity of the sensorimotor cortex during a motor task. In the tic-freely session, participants with TS showed reduced modulation of beta oscillations during movement execution compared with controls. When someone makes a hand movement, it typically comes with a decrease in beta oscillations over the contralateral sensorimotor cortex. This phenomenon is called beta desynchronization. In the current study, the modulation of beta oscillations during the task was negatively associated with tic severity, such that individuals with the most severe tics showed the least modulation of beta power. However, in the tic suppression condition, beta modulation during movement execution was not different across groups. These findings suggest that tic suppression may operate by normalizing sensorimotor beta oscillations.

ARE THE SAME MECHANISMS INVOLVED IN RESPONSE INHIBITION AND VOLUNTARY TIC SUPPRESSION?

By contrasting voluntary tic suppression with free ticcing, some of the studies mentioned above allow a better understanding of



One of the earliest studies to suggest parallels between response inhibition and voluntary tic suppression was conducted by [Serrien et al. \(2005\)](#) who investigated patterns of

EEG coherence during tic suppression and during a Go/NoGo task. In nine adults with TS, enhanced coherence within the alpha-frequency band was found across pairs of electrodes spanning the frontal cortex. In the same pairs of electrodes, TS adults' alpha coherence was increased for the NoGo trials of the Go/NoGo task relative to Go trials, while this difference was not found in control participants. It was thus concluded that the same network recruited for response inhibition during the Go/NoGo task may be engaged in voluntary tic suppression. More than a decade later, [van der Salm et al. \(2018\)](#) used a different approach to tackle a similar question. They compared brain activation in adults with TS suppressing ocular tics to that of non-TS controls suppressing eyeblinks. The main takeaway from that study is that different brain regions are recruited for eyeblink suppression in controls and tic suppression in individuals with TS. In controls, suppression of eyeblinks led to activation of brain regions commonly involved in response inhibition, such as the right ventrolateral prefrontal cortex and bilateral supplementary motor areas. In individuals with TS, tic

suppression led to increased activation in the anterior cingulate cortex and medial prefrontal regions. Thus, those latter results contrast those of Serrien et al. (2005) and suggest that the mechanism of voluntary tic suppression may be different than that of response inhibition.

Most neurocognitive studies also reported poor concordance between response inhibition and voluntary tic suppression. For instance, neuropsychologic measures of response inhibition, including the CPT as well as Go/NoGo, Flanker, Stroop, and stop-signal tasks, were not predictive of voluntary tic suppression capacities (Ganos et al., 2014c; Specht et al., 2014; Sturm et al., 2021). Similarly, Woods et al. (2008) reported that omissions errors during the CPT, but not commission errors, were associated with voluntary tic suppression (although in a small sample of 13 children). While not directly related to tic suppression, findings from behavioral therapy trials may also be relevant to this topic. Indeed, voluntary tic suppression is sometimes thought to underlie the effectiveness of behavioral therapies for TS, such as CBIT, and both tic suppression and CBIT have been shown to share some mechanisms (Morand-Beaulieu et al., 2023a). In large randomized controlled trials of CBIT in children (Chang et al., 2018) and adults (Abramovitch et al., 2017), neuropsychologic measures of response inhibition

were unrelated to CBIT outcome. However, please note that a more recent but smaller study found a link between CBIT outcome and a specific measure related to cognitive control and switching (McGuire et al., 2022). All in all, these studies suggest that the association between response inhibition measured through neuropsychologic tests and tic suppression (either voluntary or through CBIT) is not that straightforward.

CONCLUSION AND FUTURE DIRECTIONS

Over the last few decades, response inhibition has been the subject of many studies in TS, in part because of its putative role in tic suppression. Evidence supporting this hypothesis has been mixed. A possible explanation for inconsistencies across studies may be the heterogeneous profile of symptoms in individuals with TS, which complicates most studies investigating inhibitory control of voluntary actions. Another issue is that some tasks may not be sensitive enough to detect the specific difficulties in inhibitory control affecting people with TS. As shown in recent studies, very specific aspects of inhibition seem to differ in TS (Rawji et al., 2020; Indrajeet et al., 2022). The main conclusions of this chapter are outlined in Table 21.1.

Table 21.1
Summary of the literature on inhibitory control and voluntary tic suppression.

Domain	Main findings
<i>Response inhibition</i>	
Behavioral measures	• Mixed findings but a meta-analysis revealed mild difficulties with volitional inhibition in Tourette syndrome. • These difficulties are enhanced in those with co-occurring conditions and high tic severity. • Potential difficulties with automatic inhibition as well.
Neurobiologic correlates	• Increased frontal activation/functional connectivity during the inhibition of prepotent responses. • It is not clear whether those increased frontal activation/functional connectivity represent compensatory mechanisms.
<i>Voluntary tic suppression</i>	
Phenomenology	• Voluntary tic suppression may differ from other types of tic suppression, such as the reduced expression of tics in social contexts. • Some tics may be easier to voluntarily suppress than others.
Behavioral measures	• Voluntary suppressing tics may impair performance in external tasks. • Voluntary tic suppression is less efficient when focusing on external tasks.
Neurobiologic correlates	• Increased activation of several cortical regions including notably prefrontal areas during voluntary tic suppression.
<i>Parallels between response inhibition and voluntary tic suppression</i>	
Behavioral measures	• No associations between experimental measures of response inhibition and objective tic suppressibility.
Neurobiologic correlates	• Mixed findings, too few studies to draw conclusions.

In the end, this chapter comes to a similar conclusion to that of Kurvits and Ganos (2022). There is not, to this date, strong evidence supporting the relationship between the mechanisms of voluntary tic suppression and inhibitory control of voluntary actions. If tic suppression genuinely involves the same mechanisms as response inhibition, it is possible that lab tasks designed to assess response inhibition lack the necessary ecologic validity to correctly identify that effect. Furthermore, the emergence of tics and premonitory urges does not compare to a stream of stimuli in a Go/NoGo task. People suppressing their tics are not monitoring the emergence of premonitory urges and making the split-second decision to tic or not. Rather, they most likely enter a different neurocognitive state that may involve both concentration and relaxation. Therefore, future studies aiming to assess whether response inhibition, or any other psychologic process, are involved in tic suppression should do so by using an experimental design that parallels more closely the experience of tic suppression.

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